

K909 — N1 wave-2 CLOSE (all three lanes in). The partition LINE is now a FORCED THEOREM: color placement (Lyra F693, L1–L3 airtight). Colored $\rightarrow$ interior (Schur, L1) $\rightarrow$ normalizable only above the Wallach threshold (EHW, L2) $\rightarrow$ ground state = threshold rung $k=3$ (energy-monotonicity, L3, closes the K905 gap). So color rigorously decides interior-pinned (bucket-1-eligible) vs boundary-free (bucket-2 modulus) — the partition-boundary mechanism, EARNED. Residual correctly held: $a=N_c$ is identification-tier (L4); the mass-READING (Conjecture C) is a separate open lemma. The wave splits the common root (K908) exactly: the WHERE is forced (placement), the WHAT is open (mass-species/CP-phase). Net of wave 2: TWO disciplined negatives (Conjecture C not forced — Grace; CP-exists re-asserts domain-complex — Elie, self-retraction) + ONE genuine new theorem (the partition line = color) + a sharper architecture (genus/species). Zero false promotions; guardrail held on all three.

Keeper | 2026-07-25 Sat | Wave 2 closed. The boundary between the buckets is proved — and it's color. The values inside bucket 1 remain row-by-row. Honest, and a real advance.

The three lanes (all accepted)

Lyra F693 — the partition LINE is FORCED (placement theorem, ACCEPTED)

- L1 colored $\rightarrow$ interior: Schur — boundary-trace is color-equivariant, Shilov boundary carries only color-singlets $\rightarrow$ colored state has identically zero boundary overlap $\rightarrow$ interior-supported. AIRTIGHT.
- L2 min normalizable weight = Wallach threshold $k=3$: EHW/Rossi–Vergne; $k=1,2$ non-normalizable (electron $k=1$ is a boundary state). AIRTIGHT.
- L3 ground state = threshold: conformal energy = weight k , strictly increasing on admissible $\{k \geq 3\} \rightarrow$ minimizer = $k=3$, uniquely. Two exclusions: $k=1,2$ by non-normalizability, $k \geq 4$ by energy. (Casimir ties $k=2,3$ at -6 ; normalizability breaks the tie $\rightarrow$ L2 load-bearing. Verified.) AIRTIGHT — closes the K905 “why the ground rung” gap.
- L4 $a=N_c$: held IDENTIFICATION-TIER (the threshold integer equaling the color number, banked T2511/F96/K313), NOT folded into airtight. Correct.
- $\square$ the partition LINE is a THEOREM: color decides interior-pinned (colored $\rightarrow$ bucket-1-eligible) vs boundary-free (colorless $\rightarrow$ bucket-2 modulus). The elegant heart of the partition (K907 reframe), earned. Modulo the $a=N_c$ identification.

Grace — Conjecture C NOT FORCED (mass-species open, ACCEPTED, K908)

BT gives “mass= $1/\varepsilon$ ” a home + right-sign down match, but doesn't PICK $1/\varepsilon$ (four mass operators across sectors; $1/\varepsilon$ in one). Quark-mass VALUES stay bucket-1-pending.

Elie — CP-exists RE-ASSERTS domain-complex (CP-phase-species open, ACCEPTED, K908)

Toy 4857: J splits by the STATE not the domain; real radial states $\rightarrow J=0$; F498 gates it. Self-retracted 4854. CP-exists stays bucket-1-pending.

The clean split of the common root (K908 → resolved)

K908 posed: both open rows hinge on “is the condensate structure forced?” Lyra’s placement theorem SPLITS it precisely: - The WHERE — placement (which masses are bucket-1-eligible): FORCED (color, Lyra F693). - The WHAT — the value (the mass functional / the CP phase): OPEN (Conjecture C mass-species; the $\mathbb{Z}_3$ -phase for CP). So Lyra forced the placement (the boundary), not the symbol/phase (the values). The two open rows now share ONE remaining gate: is the condensate’s SYMBOL (mass profile) and PHASE (CP) forced onto the placed mode? — the mass-species (Conjecture C) + the CP-phase (Engine B), both still open.

Net N1 state after wave 2 (the honest capstone, grown)

- Containment (genus): THEOREM (F595) — every derived observable is SOME functional of μ .
- Bucket 2 (moduli): characterized (lepton masses K902, mixing sizes, CP magnitude, m_u loose-trap).
- ★ Partition LINE (which are bucket-1-eligible vs bucket-2): FORCED THEOREM (color placement, Lyra F693). NEW this wave. Modulo $a=N_c$ identification.
- Bucket-1 SPECIES (values are forced functionals): ROW-BY-ROW, candidate (Conjecture C mass-species, CP-phase both open; lives-there for mixings pending).
- Honest publishable result NOW: the FRAMEWORK + containment theorem + characterized moduli + the FORCED color partition-line. That is Lyra’s “sharp, proven map of which parameters the geometry pins vs leaves free, WITH a proved mechanism (color) for the boundary.” A genuine theorem-grade result — NOT “the SM is derived.”
- The horizon (strong theorem): every bucket-1 row’s value a forced functional — row-by-row, gated on the species (condensate symbol + phase forced).

Discipline note

Wave 2 = the push working as designed: two clean negatives (one a self-retraction), one earned theorem, zero false promotions. The genus/species line (K908) held every promotion honest — the partition LINE promoted (mechanism forced), the VALUES did not (species not forced). The $a=N_c$ and Conjecture-C riders were correctly kept OUT of “airtight.” Peak-convergence guardrail intact.

Directions (wave 3 — the shared gate)

- ★ THE ONE REMAINING GATE for both open bucket-1 rows: is the condensate SYMBOL (mass profile, Conjecture C) + PHASE (CP, Engine B) FORCED onto the placed mode? The placement is forced (Lyra); the symbol/phase is the last open thing. Grace (mass-species) + Elie (CP-phase) re-fire together on the condensate structure; Lyra sources whether the ν_R condensate profile+phase is geometry-forced or free.
- ★ LYRA (Task #1): fold F693 into the architecture — the capstone now includes a FORCED partition-line theorem; state genus/species; the strong theorem is row-by-row on the species.
- ★ LYRA — lives-there (Task #6): the mixing-values species (parallel row-promotion).
- KEEPER: partition-line theorem accepted (color, modulo $a=N_c$); bucket-1 species held candidate; the condensate symbol+phase is the shared wave-3 gate; genus/species is the standing bucket-1 test.

— Keeper K909, 2026-07-25. Wave-2 close: the partition LINE is a FORCED theorem (Lyra F693 color placement, L1–L3 airtight; closes K905’s lowest-rung gap; $a=N_c$ held identification-tier). Grace (Conjecture C not forced) + Elie (CP re-asserts domain-complex, self-retraction) = mass-species + CP-phase stay open. The WHERE is forced (placement/boundary = color), the WHAT is open (values = condensate symbol+phase). Capstone grown: FRAMEWORK + containment theorem + moduli + FORCED color partition-line = publishable; strong theorem row-by-row. Wave-3 shared gate: is the condensate symbol+phase forced? See [\[\[Keeper_K908_wave2_two_clean_negatives\]\]](#), [\[\[Lyra_F693_lowest_rung_link_airtight\]\]](#), [\[\[Keeper_K907_wave1_consolidation\]\]](#), [\[\[Keeper_K906_N1_DEEP_PUSH_CHARTER\]\]](#).